When does drinking become a pattern?

Is it possible to predict student alcohol consumption from school and social context?

Team 24

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Abstract

Alcohol drinking habits among the secondary school students are a complex behavioral In fact, the phenomenon is affected by academic, social as well as family-related This paper investigates the possibility of utilizing the results of structured surveys to predict the number of weekday alcohol consumption levels using a cost-sensitive machine learning framework. The comparison of several multi-class models based on varying definitions of targets, class imbalance problems and feature subsets.

Results underscore the natural difficulty in the task, maintained by the weakly informative predictors and strong overlap between neighboring consumption levels. Variables related to school arise as the strongest contextual predictor of test performance, while such variables account for the largest amount of variance

Social and family characteristics have little discriminatory power. The comparison between the three-class and five-class formulations shows there is a trade-off between them. between Predictive Stability and Semantic Resolution. Although class aggregation reduces the values of agreement metrics, five-class models by oversampling more accurately preserves its ordinal fidelity and obtains competitive results for cost-sensitive performance. In general, the results indicate the viability of oversampling five-class models for detailing risk stratification in educational environments.

Introduction

Alcohol consumption among secondary school students is a well-documented phenomenon with potential implications for academic performance, health and social development.

Understanding the factors associated with such behaviors is therefore of interest from both an educational and a societal perspective.

This project investigates whether alcohol consumption levels can be inferred from observable characteristics related to a student's academic environment and family background. The research question driving the analysis is defined as follows:

Is it possible, by analyzing school-related or family and social-related attributes, to predict the alcohol consumption of a student?

Rather than focusing on causal inference, the objective is to evaluate the predictive potential of structured survey data through a systematic machine learning pipeline. This report documents the full workflow, from data acquisition and exploratory analysis to predictive modeling and comparative evaluation.

Datasets

The analysis is based on the *Student Alcohol Consumption* dataset, publicly available on Kaggle[1] and originally collected from Portuguese secondary schools. The data consist of two separate but structurally identical files:

- `student-mat.csv`, referring to students enrolled in the Mathematics course (396 records)
- `student-por.csv`, referring to students enrolled in the Portuguese language course (650 records)

Both datasets share the same set of attributes and were collected through standardized questionnaires aimed at capturing students' academic performance, family background and social behavior. Each record corresponds to a single student and includes variables describing demographic characteristics, parental education and occupation, school-related factors and lifestyle habits.

[1] All the dataset are availabe to this link: https://www.kaggle.com/uciml/student-alcohol-consumption

Variables Description

The dataset contains a total of 33 attributes describing students' demographic characteristics, family background, academic context and lifestyle habits. Variables are a mix of categorical, ordinal and numerical features, encoded according to standardized questionnaire scales.

For clarity, the attributes are grouped into the following categories.

Demographic and school-related variables. These variables describe basic personal information and the academic environment:

- `school`: student's school (`GP`, `MS`);
- `sex`: gender (`F`, `M`);
- `age`: age in years (15–22);
- `address`: home address type (`Urban`, `Rural`);
- `famsize`: family size (`LE3` $\leq$ 3 members, `GT3` $>$ 3 members);
- `Pstatus`: parents' cohabitation status (`Together`, `Apart`).

Family background variables. These features capture parental education, occupation and household conditions:

- `Medu`, `Fedu`: mother's and father's education level (0–4);

– `Mjob`, `Fjob`: mother's and father's occupation;
– `guardian`: student's legal guardian;
– `famrel`: quality of family relationships (1–5).

Academic and support-related variables. These variables describe study habits, school support and academic history:

– `traveltime`: home-to-school travel time (1–4);
– `studytime`: weekly study time (1–4);
– `failures`: number of past class failures (0–3);
– `schoolsup`, `famsup`: educational support indicators (`yes/no`);
– `paid`: extra paid classes (`yes/no`);
– `activities`: extra-curricular activities (`yes/no`);
– `nursery`, `higher`, `internet`: educational and infrastructure indicators (`yes/no`);.
– `G1, G2, G3`: academic outcomes per period (respectively first period, second period and final grade) (0–20);

Lifestyle and social behavior variables. These attributes describe social activities, health perception and free-time habits:

– `romantic`: romantic relationship status (yes/no);
– `freetime`: free time after school (1–5);
– `goout`: frequency of going out with friends (1–5);
– `health`: self-reported health status (1–5).

Alcohol consumption variables. Alcohol intake is measured through two ordinal variables:

– `Dalc`: weekday alcohol consumption (1–5);
– `Walc`: weekend alcohol consumption (1–5).

Data integration, preprocessing and EDA

The original datasets do not provide a unique student identifier. As a consequence, integrating the Mathematics and Portuguese datasets requires a careful matching strategy to reliably identify students enrolled in both courses.

To minimize the risk of incorrectly merging records belonging to different individuals, students are matched using a combination of shared attributes describing demographic characteristics, family structure and educational context. This multi-attribute comparison significantly reduces the probability of false matches.

Students enrolled in both courses are consolidated into a single observation. A new variable representing the number of attended courses is introduced. For these students, course-specific variables related to academic outcomes and attendance are aggregated by computing mean values for absences and grade-related variables.

As part of the preprocessing phase, all variables originally encoded as strings are transformed into numerical representations, ensuring compatibility with downstream statistical and machine learning models.

Additional preprocessing steps include:

– verification and correction of variable data types
– consistency checks across correlated attributes

No imputation or aggressive filtering is applied in order to preserve the original data distribution.

The exploratory data analysis (EDA) was conducted to gain an initial understanding of alcohol consumption patterns among students and to investigate their relationship with social, academic and family-related factors. At this stage, the analysis remains purely descriptive and does not introduce any predictive modeling assumptions.

The EDA follows a structured approach, combining univariate analyses of alcohol consumption variables with bivariate analyses based on box plots conditioned on contextual factors. All visualizations are generated directly from the preprocessed data, without applying model-driven transformations, in order to preserve the original semantics of the variables and avoid information loss.

Distribution of Alcohol Consumption

Alcohol intake is measured through two ordinal variables: `Dalc`, representing weekday alcohol consumption and `Walc`, representing weekend alcohol consumption, both defined on a discrete scale from 1 (very low) to 5 (very high).

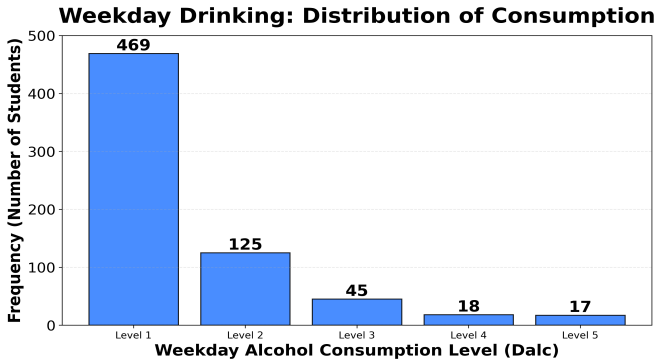

Figure 1: Distribution of alcohol consumption during weekdays (`Dalc`).

Figures 1 and 2 illustrate the distribution of alcohol consumption during weekdays and weekends, respectively, highlighting a clear contextual difference.

Weekday alcohol consumption is strongly concentrated at the lowest level, with the vast majority of students reporting minimal or no alcohol intake during the school week. Higher weekday consumption levels occur only sporadically, indicating that alcohol use is not a routine daily behavior.

In contrast, weekend alcohol consumption exhibits a markedly more dispersed distribution. Although lower consumption levels remain prevalent, moderate and high values occur substantially more frequently, suggesting that alcohol use is largely concentrated during weekends and social occasions.

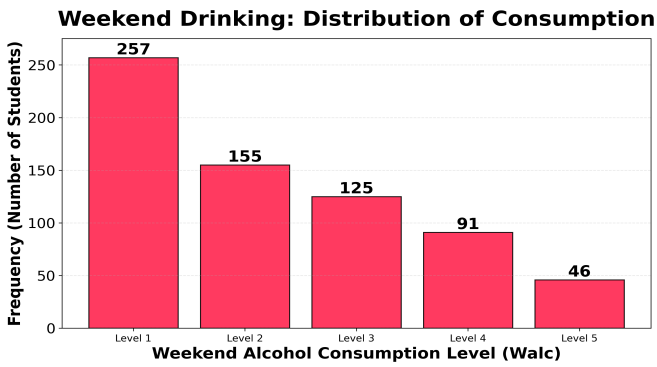

Figure 2: Distribution of alcohol consumption during weekends (`Walc`).

This discrepancy emphasizes the context-dependent nature of alcohol consumption among students and motivates the analysis of social, academic and family-related factors.

Social Activity and Alcohol Consumption

Social behavior is captured by the variable `goout`, which measures how frequently students go out with friends. Figures 3 and 4 report weekend and weekday alcohol consumption conditioned on social activity levels, respectively.

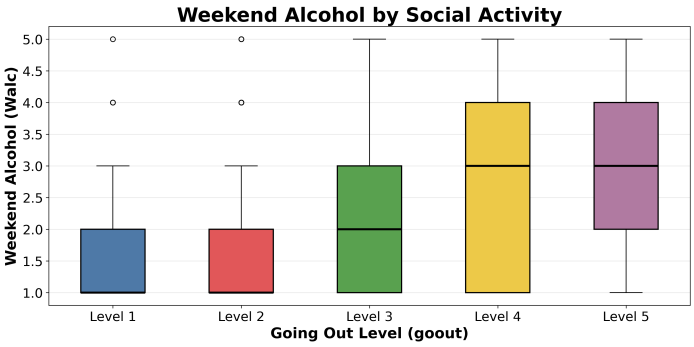

Figure 3: Weekend alcohol consumption (`Walc`) by level of social activity (`goout`).

A clear increasing trend emerges for weekend consumption: students who report more frequent social outings tend to exhibit higher median alcohol intake and greater variability, indicating that alcohol use is strongly embedded within social contexts.

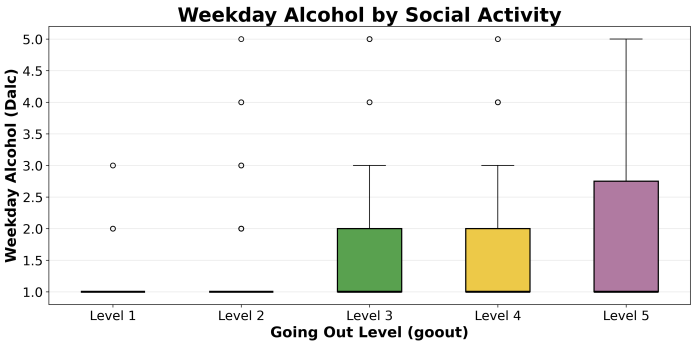

Figure 4: Weekday alcohol consumption (`Dalc`) by level of social activity (`goout`).

In contrast, weekday alcohol consumption remains consistently low across all levels of social activity, reinforcing the interpretation that alcohol use is primarily associated with weekend social interactions.

Academic Performance and Weekday Alcohol Consumption

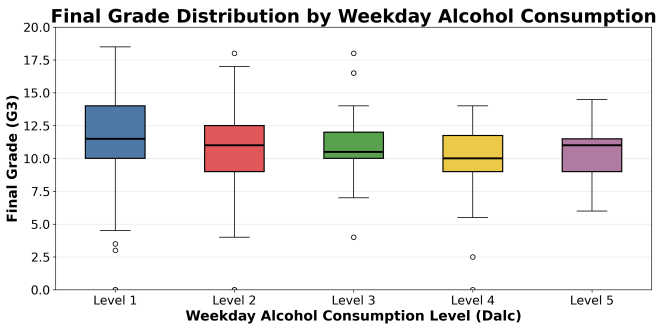

Figure 5: Final grade distribution (`G3`) by weekday alcohol consumption level (`Dalc`).

Academic performance is analyzed through the final grade variable `G3`. Figure 5 reports the distribution of final grades conditioned on weekday alcohol consumption levels.

Lower weekday alcohol consumption is associated with slightly higher median final grades and a broader upper tail. As consumption levels increase, median grades tend to decrease modestly, although substantial overlap between distributions persists across all levels. This overlap suggests that weekday alcohol intake alone is insufficient to explain academic performance and that its effect is likely mediated by additional academic, social and family-related factors.

Family Relationships and Alcohol Consumption

Family context is explored using the variable `famrel`, representing the quality of family relationships.

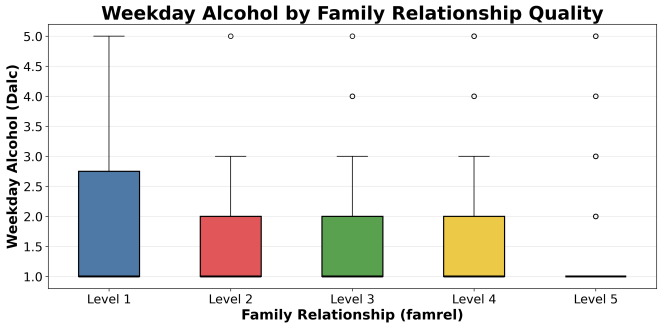

Figure 6: Weekday alcohol consumption (`Dalc`) by family relationship quality (`famrel`).

Figure 6 reports weekday alcohol consumption conditioned on family relationship quality.

Stronger family relationships are generally associated with lower median alcohol consumption and reduced variability. However, the effect is less pronounced than that observed for social activity, suggesting a secondary but non-negligible role of family context.

Overall, the exploratory analysis highlights alcohol consumption as a multifactorial phenomenon, characterized by strong social components and weaker, more diffuse associations with academic performance and family relationships. The extensive overlap observed across all bivariate distributions indicates that no single variable is sufficient to explain consumption patterns in isolation, motivating the use of multivariate and non-linear modeling approaches in the subsequent predictive analysis.

Categorical variables were binarized using one-hot encoding only in the modeling phase, after the completion of the exploratory analysis, to preserve interpretability during the descriptive stage.

Modeling Strategy and Design Choices

The modeling phase focuses on predicting alcohol consumption using the variable `Dalc`, which measures weekday drinking behavior. This choice was driven primarily by the interpretative meaning of the variable rather than by purely statistical considerations.

High alcohol consumption during weekends may reflect episodic or socially contextualized behavior and does not necessarily indicate problematic or persistent drinking patterns. In contrast, weekday alcohol consumption is more likely to capture habitual behaviors that interfere with daily routines and re-

sponsibilities, making it more informative from a behavioral and educational perspective.

Although the `Dalc` distribution exhibits a strong class imbalance, it was intentionally retained as the target variable due to its higher semantic relevance. Modeling weekday consumption allows the analysis to focus on patterns that are less influenced by temporary social contexts and more closely associated with underlying lifestyle and academic functioning.

The predictive task was initially formulated as a multi-class classification problem with five ordered classes, corresponding to increasing levels of alcohol consumption. This fine-grained formulation was chosen to preserve as much information as possible and to explicitly model subtle differences between adjacent consumption levels.

However, empirical results revealed substantial limitations in the five-class setting. Severe class imbalance, combined with strong overlap between neighboring categories, led to unstable predictions, low inter-class agreement and poor generalization performance, particularly for extreme consumption levels. To address these issues, the prediction task was reformulated into a reduced three-class setting by aggregating adjacent categories according to the following mapping: the original classes 1 and 2 were preserved as distinct categories, while classes 3, 4 and 5 were merged into a single high-consumption class. This transformation resulted in a more stable learning problem, improving predictive reliability and interpretability while retaining the most relevant distinctions for downstream analysis.

An additional methodological challenge arises from the inherently ordinal nature of the target variable. Alcohol consumption levels exhibit a natural ordering, yet standard multi-class classification models treat classes as nominal and equidistant, ignoring ordinal structure and potentially penalizing small, semantically mild errors as severely as large missclassifications. This limitation motivated the exploration of modeling strategies capable of implicitly accounting for ordering effects.

Class imbalance represented a central challenge throughout the analysis, particularly in the original five-class formulation. To assess its impact on model performance, two alternative strategies were considered: stratified cross-validation and oversampling.In the first case, models were trained and evaluated using stratified cross-validation with a fixed random seed, preserving the original class distribution while ensuring balanced folds. In the second case, oversampling was applied to the training data to rebalance the class distribution and improve sensitivity to minority classes.This comparative setup allows the analysis to disentangle the effects of data rebalancing from those of model choice, providing insight into which strategy yields more robust and reliable performance under class imbalance.

To disentangle the contribution of different contextual factors, the analysis was conducted using two distinct feature subsets. The first focuses on **school-related attributes**, including the following attributes `school`, `sex`, `age`, `address`, `traveltime`, `studytime`, `activities`, `higher`, `internet`, `freetime`, `goout`, `health`, `absences`, `G1`, `G2`, `G3`, `N_courses`, `paid`, `Dalc`, `Walc`.

The second concentrates on the **social and family dimension**, selecting variables related to demographic information, household structure, parental background, family relationships, leisure activities and social behavior(the following attribute where used:`school`, `sex`, `age`, `address`, `famsize`, `Pstatus`, `Medu`, `Fedu`, `Mjob`, `Fjob`, `guardian`, `traveltime`, `schoolsup`, `famsup`, `activities`, `nursery`, `internet`, `romantic`, `famrel`, `freetime`, `goout`, `health`, `Dalc`,`Walc`)

After data partitioning, a correlation analysis was performed within each subset to identify strongly redundant predictors. In the school-related feature set, the variables `G1` and `G2` were excluded from the modeling phase, as they exhibit a strong correlation with the final grade `G3` and would otherwise introduce unnecessary redundancy and information leakage.

In the modeling phase, Weekend Alcohol Consumption (`Walc`) was retained as a predictive feature for Weekday Alcohol Consumption (`Dalc`). This choice is grounded in the distinction between two different behavioral patterns: recreational social drinking (typically occurring on weekends) and habitual/problematic consumption (occurring during the week). The moderate correlation observed between the two variables confirms that while related, they represent distinct phenomena: a high `Walc` does not automatically imply a high `Dalc`. By including it, the model is tasked with a more complex objective: identifying which environmental factors (school-related or family-related) act as catalysts, distinguishing students who limit consumption to leisure time from those whose consumption extends into the academic week. Therefore, `Walc` serves as a behavioral baseline rather than a direct proxy.

Additionally, during model development, the numerical variables `absences` and `G3` were normalized to account for scale differences and to stabilize learning for models sensitive to feature magnitude.

These preprocessing considerations were not required for the social–family feature set, where variables are predominantly categorical or ordinal and do not exhibit strong linear dependencies. Overall, this separation enables a controlled comparison between academic and social–family contexts under identical modeling and evaluation conditions, ensuring that observed performance differences reflect contextual information rather than modeling artifacts.

Model Selection and Parameter Configuration

All the classification models adopted in this study were deliberately selected among **natively multi-class algorithms**, in order to directly address the multi-level structure of alcohol consumption without relying on external class decomposition strategies (e.g., one-vs-all or one-vs-one schemes). This design choice ensures that class probabilities are learned jointly and coherently, thereby improving predictive stability and consistency in the presence of class imbalance.

Given the heterogeneous nature of the predictors and the limited sample size, particular emphasis was placed on robustness, generalization ability and interpretability. To this end, all classification models were trained within an **Attribute Selected Classifier** framework, ensuring that feature selection was performed exclusively on the training folds in order to prevent information leakage, feature selection was performed using a CfsSubsetEval evaluator with BestFirst search. An exception was made for the ordinal model, which incorporates an internal mechanism for attribute selection.

A cost-sensitive classifier was not adopted, as such approaches explicitly bias the training process toward predefined misclassification costs, potentially overshadowing the contribution of the original feature space, which was the primary object

of investigation in this study.

CART-Based Decision Tree Models

As interpretable baselines, CART-based decision tree models were employed using the `SimpleCart` algorithm. SimpleCart natively supports multi-class classification and provides transparent decision rules, enabling direct inspection of the relationships between socio-demographic, family-related and school-related variables.

To ensure stable and easily interpretable trees, model complexity was deliberately constrained. A minimum of 10 instances per leaf was required, preventing the creation of fragile branches driven by small sample fluctuations. Cost–complexity pruning was applied using 5-fold internal cross-validation, with the complexity parameter set to 0.8 in order to limit excessive tree growth while preserving relevant splits. Binary splits were enforced for nominal attributes, while midpoint splits were adopted for numerical variables, ensuring consistent and robust partitioning of the feature space. A static random seed was fixed to guarantee full reproducibility of the baseline results.

Logistic Model Trees

To bridge the gap between interpretability and predictive performance, **Logistic Model Trees (LMT)** were included as an advanced hybrid model. LMTs combine a hierarchical decision tree structure with local multinomial logistic regression models at the leaves, allowing class probabilities to be modeled smoothly while preserving an interpretable decision process.

Model complexity was explicitly controlled to enhance robustness under limited sample size and class imbalance. The maximum number of logistic optimization iterations was set to 50 to ensure convergence of the local regression models, while a minimum of 15 instances per leaf was enforced to reduce instability in terminal nodes. In addition, ridge regularization with a coefficient of 0.1 was applied to the logistic models in order to mitigate the effects of correlated predictors and prevent overfitting. All experiments were conducted using a fixed random seed to ensure reproducibility.

Random Forest

As the primary performance-oriented classification model, **Random Forests** were employed. Random Forests are natively multi-class and particularly well-suited to heterogeneous feature spaces, offering strong robustness to noise, multicollinearity and non-linear interactions.

The ensemble was composed of 100 decision trees, a choice aimed at reducing variance and improving stability without increasing model aggressiveness. At each split, 6 candidate features were randomly selected, providing a balanced trade-off between tree diversity and split informativeness. To further control model complexity, the maximum depth of individual trees was limited to 5, preventing excessively deep structures and reducing the risk of overfitting. Training was performed using a single execution thread and a static random seed was fixed to guarantee full reproducibility of the experimental results.

Gradient Boosted Trees for Ordinal Modeling

In addition to the multi-class classification models, **Gradient Boosted Trees** were trained in a **regression setting** to explicitly account for the ordinal nature of the target variable. Rather than treating alcohol consumption levels as purely categorical outcomes, this approach models consumption as a continuous latent variable, directly exploiting the inherent ordering of the classes.

The squared-error loss function was adopted, ensuring that larger ordinal prediction errors are penalized more severely than smaller ones and aligning the learning objective with the semantic structure of the target variable. This regression-based formulation provides a complementary perspective to standard classification models and serves as a robustness check for the observed performance trends. As for the other models, a fixed random seed was used to ensure reproducibility.

Overall, the modeling strategy aims to strike a balance between interpretability, robustness and predictive performance. Multi-class classification tasks were handled using models that natively support multiple classes, while the ordinal nature of the target variable was investigated through a regression-based Gradient Boosted Tree approach. Feature selection was embedded within the training process and no external class decomposition techniques were adopted, resulting in a pipeline that is coherent with both the data structure and the learning objectives.

Evaluation Metrics

Given the multi-class nature of the problem, evaluation metrics were selected to properly account for class imbalance and to provide a fair assessment of performance across all consumption levels. Let n denote the total number of classes. Both the F1-score and the Balanced Accuracy were computed using a macro-averaging strategy, assigning equal importance to each class regardless of its frequency.

For each class $k \in \{1, \ldots, n\}$, precision and recall are defined as:

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \qquad \text{Recall}_k = \frac{TP_k}{TP_k + FN_k},$$

where TP_k, FP_k and FN_k denote the number of true positives, false positives and false negatives for class k, respectively.

The class-specific F1-score is then computed as the harmonic mean of precision and recall:

$$\text{F1}_k = \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}.$$

To obtain a global measure of performance, the macro-averaged F1-score is defined as:

$$\text{F1}_{\text{macro}} = \frac{1}{n} \sum_{k=1}^{n} \text{F1}_k.$$

This formulation ensures that poor performance in minority classes is not masked by strong results in majority classes.

Balanced Accuracy is defined as the macro-average of class-wise recall values:

$$\text{Balanced Accuracy} = \frac{1}{n} \sum_{k=1}^{n} \text{Recall}_k.$$

This metric directly measures the model's ability to correctly identify instances of each class and is particularly suitable in multi-class, unbalanced settings, where standard accuracy may be misleading.

In addition to standard performance metrics, a cost-sensitive evaluation was performed to explicitly account for the ordinal structure of the target variable. A cost matrix was defined such that the cost associated with predicting class $\hat{y}$ when the true class is y is proportional to the square of absolute distance between the two classes:

$$C(y, \hat{y}) = (y - \hat{y})^2$$

The total cost of missclassification is computed as follows.

$$\text{Total Cost} = \sum_{i=1}^{N} C(y_i, \hat{y}_i)$$

where N denotes the number of observations, y_i the true class and $\hat{y}_i$ the predicted class for the i-th instance.

Five classes

True	Prediction				
	1	**2**	**3**	**4**	**5**
1	0	1	4	9	16
2	1	0	1	4	9
3	4	1	0	1	4
4	9	4	1	0	1
5	16	9	4	1	0

Three classes

True	Prediction		
	1	**2**	**3**
1	0	1	4
2	1	0	1
3	4	1	0

Table 1: Cost matrices for the five-class and three-class.

The table above illustrate the cost structures adopted for the five and three class settings, respectively. The squared formulation of the cost function was adopted to emphasize the ordinal nature of the target variable. By using the square of the distance between classes, larger penalties are assigned to severe misclassifications involving distant classes, while comparatively mild errors between adjacent classes are penalized less heavily. This design choice reflects the intuition that confusing neighboring consumption levels is substantially less critical than predicting extreme categories and it allows the evaluation to better capture the practical relevance of prediction errors.

ROC curves and lift charts were not employed in this analysis, as their interpretation becomes less straightforward in a multi-class setting and would be further complicated by the simultaneous comparison of four different models. Similarly, no formal statistical tests between models were conducted, since such tests are typically designed for pairwise comparisons, whereas the present study focuses on the comparative evaluation of multiple models rather than on hypothesis testing between two alternatives.

Model Evaluation and Comparative Analysis

Once the evaluation framework was defined, all trained models were assessed under identical experimental conditions to ensure a fair and consistent comparison. For each experimental configuration, the evaluation metrics introduced in the previous section were computed at fold level and subsequently aggregated to obtain mean performance estimates.

The evaluation results were consolidated into a unified comparison table reporting Balanced Accuracy, macro F1-score and Cohen's Kappa and the total missclassification cost derived from the predefined cost matrix. This consolidated representation enables a comprehensive comparison across heterogeneous models, capturing not only predictive performance but also robustness to class imbalance, inter-class agreement and the severity of ordinal missclassifications.

Performance Comparison on Social–Family Analysis

Figures 7 and 8 present the comparative performance of the evaluated models when the feature space is restricted to social and family-related variables, considering both five-class and three-class target formulations under stratified cross-validation and oversampling-based rebalancing.

In the five-class setting (Figure 7), predictive performance remains generally limited across all models. Balanced Accuracy values are low and show moderate variability depending on the rebalancing strategy, while Cohen's Kappa consistently indicates weak inter-class agreement. These results underline the intrinsic difficulty of discriminating among fine-grained alcohol consumption levels using exclusively relational attributes, where adjacent ordinal classes exhibit substantial overlap. While oversampling improves Balanced Accuracy for some models, most notably Gradient Boosted Trees, it does not translate into a systematic improvement in agreement-based metrics, suggesting that increased sensitivity to minority classes does not necessarily yield more coherent class separation. Moreover, ensemble-based approaches do not consistently outperform simpler models, indicating that model complexity alone is insufficient to resolve the ambiguity introduced by both class imbalance and target granularity.

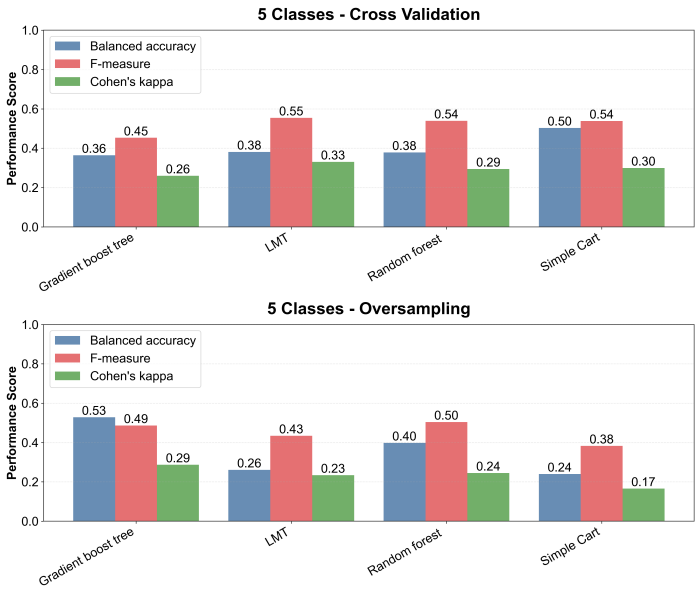

Figure 7: Model performance on social–family features in the five-class setting.

When the target is reformulated into three aggregated classes (Figure 8), a clear and systematic improvement is observed across all metrics and models. Balanced Accuracy and macro F-measure increase substantially under both stratified cross-validation and oversampling, and Cohen's Kappa values indicate more stable and consistent agreement. This improvement reflects a significant reduction in ordinal ambiguity, as class aggregation mitigates overlap between neighboring consumption levels and enables more robust decision boundaries. Nonetheless, the improvement is achieved at the expense of descriptive resolution, as distinctions between moderate and extreme consumption behaviors are partially collapsed.

The comparison between stratified cross-validation and oversampling highlights a non-trivial trade-off. Oversampling generally improves macro-averaged metrics, particularly in the three-class formulation, confirming its effectiveness in alleviating bias toward majority classes. However, gains in F-measure are not always accompanied by proportional increases in Cohen's Kappa, indicating that oversampling primarily enhances class-wise recall rather than global agreement, and may introduce additional uncertainty in class boundary definition.

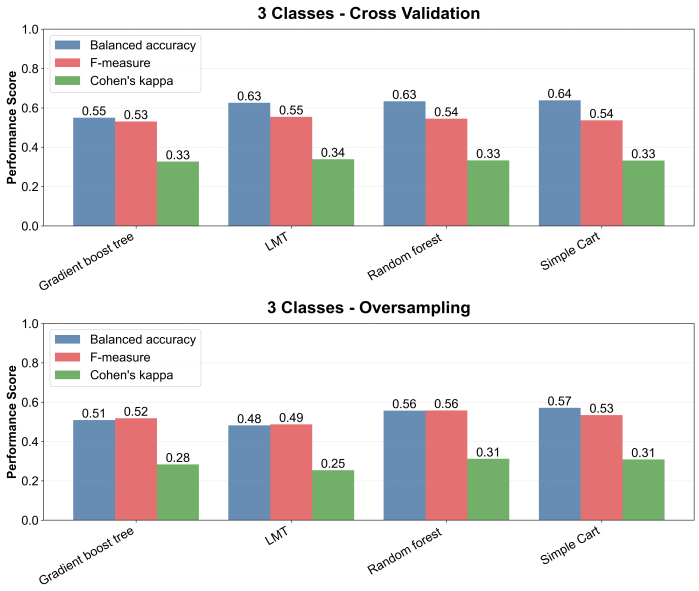

Figure 8: Model performance on social–family features in the three-class setting.

Cost-Sensitive Evaluation

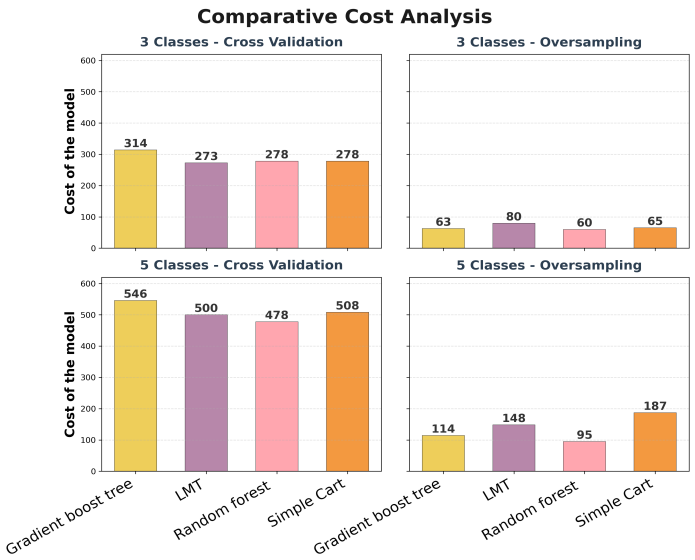

Figure 9: Total misclassification cost for social–family models.

To account for the ordinal severity of misclassification errors, model performance was analyzed using a cost-sensitive framework based on a predefined ordinal cost matrix. Figure 9 reports the total misclassification cost across models, target formulations and rebalancing strategies. Oversampling leads to a substantial reduction in total cost in both the three-class and five-class settings, confirming class imbalance as a major contributor to high-severity ordinal errors. In the three-class formulation, cost values are consistently lower across models, with Gradient Boosted Trees and Random Forest achieving the lowest observed costs, although differences remain modest. This indicates that ensemble methods are slightly more effective at limiting severe ordinal deviations without fundamentally altering task difficulty. Notably, improvements in accuracy-based metrics under oversampling do not always translate into proportional cost reductions. In the three-class setting, misclassifications become less frequent but often span larger ordinal jumps, producing a compensation effect between error frequency and magnitude. Conversely, in the five-class formulation, some models achieve relatively low cost despite weak inter-class agreement, reflecting conservative prediction strategies biased toward central classes. While effective in minimizing ordinal distance, such strategies may fail to identify extreme consumption profiles, highlighting the trade-off between risk sensitivity and ordinal error minimization.

Performance Comparison on School-Related Analysis

Figures 10 and 11 report the performance of the evaluated models when the feature space is restricted to school-related variables. Compared to the social–family domain, this setting exhibits a more structured predictive signal, resulting in consistently higher performance across models and evaluation metrics.

Figure 10: Model performance on school-related features in the five-class setting.

In the five-class formulation (Figure 10), predictive performance remains affected by the fine granularity of the target, as reflected by moderate Balanced Accuracy values and relatively low Cohen's Kappa scores. Nonetheless, models display more differentiated behavior than in the social–family analysis, indicating that school-related attributes—such as academic performance indicators and attendance patterns—provide a stronger and more coherent signal. In this setting, ensemble-based methods show a clearer advantage over simpler models, suggesting that increased model capacity partially compensates for the complexity introduced by the five-level ordinal target.

When the target is aggregated into three classes (Figure 11), a substantial improvement is observed across all evaluation metrics. Balanced Accuracy, macro F-measure and

Cohen's Kappa increase consistently under both stratified cross-validation and oversampling, reflecting improved class separability and more stable decision boundaries. Random Forest and Gradient Boosted Trees achieve the highest agreement scores under oversampling, confirming that class aggregation enables models to exploit school-related predictors more effectively. However, this gain in predictive stability comes at the cost of reduced descriptive resolution, as distinctions between intermediate and extreme consumption levels are partially obscured by the aggregation process.

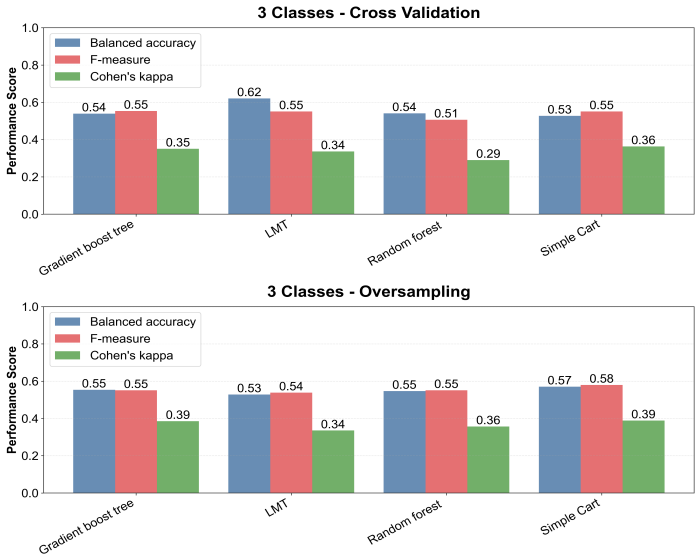

Figure 11: Model performance on school-related features in the three-class setting.

Cost-Sensitive Evaluation

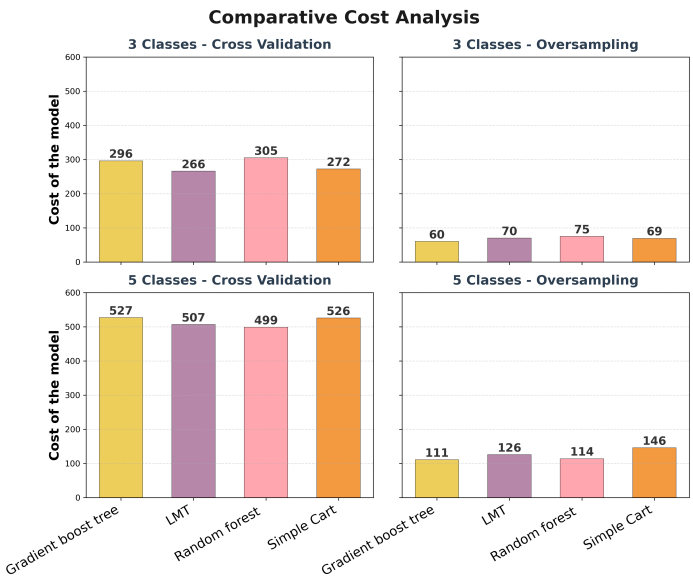

Figure 12: Total misclassification cost for school-related models.

Figure 12 summarizes the total misclassification cost for school-based models under cross-validation and oversampling. Consistent with previous findings, oversampling substantially reduces total misclassification cost, confirming class imbalance as a primary driver of high-severity ordinal errors. In the three-class formulation, Gradient Boosted Trees achieve the lowest observed cost, reflecting an effective balance between error frequency and ordinal severity after target simplification.

Conclusions

This study examined the feasibility of predicting student alcohol consumption using observable characteristics from academic, social and family contexts, formulating the task as a cost-sensitive ordinal classification problem.

The results consistently highlight the intrinsic difficulty of the prediction task, largely driven by the heterogeneous and indirect nature of the available predictors, which induce substantial overlap between adjacent consumption levels.Across all experimental settings, school-related variables emerged as systematically more informative than social–family attributes. Indicators related to academic performance, attendance and study behaviors provide a more structured and stable predictive signal, leading to higher inter-class agreement and lower ordinal misclassification costs.

In contrast, social and family features alone proved insufficient to support reliable fine-grained discrimination, particularly under the five-class formulation.

A central methodological finding concerns the trade-off between target granularity and decision reliability. Aggregating the target into three classes improves agreement-based metrics and reduces the overall frequency of misclassification. However, these gains are partially offset by an increase in the severity of individual errors, as misclassifications in the three-class setting tend to span larger ordinal distances. As a result, improvements in nominal accuracy do not translate proportionally into lower operational risk.

Conversely, the five-class formulation preserves the semantic richness of the original target, enabling a more nuanced representation of consumption patterns. Although this setting is inherently more challenging and yields lower agreement scores, the combination of oversampling and ensemble learning strategies, particularly Random Forest—allows models to achieve competitive cost-sensitive performance. In this configuration, models exhibit a conservative prediction behavior that limits severe ordinal errors while maintaining the descriptive resolution necessary to distinguish intermediate risk levels.

Overall, the school-related analysis indicates that, unlike the social–family domain, the five-class formulation remains a viable and meaningful modeling choice when supported by appropriate rebalancing and ensemble methods. While three-class models offer greater stability, the oversampled five-class configuration provides a more balanced compromise between predictive reliability and semantic fidelity.

Among the evaluated approaches, the Random Forest model trained on school-related features with oversampling demonstrates the most robust overall behavior, achieving a favorable balance between total misclassification cost, interpretability and ordinal consistency. From an applied perspective, these findings suggest that when the objective extends beyond coarse risk detection toward nuanced stratification of alcohol consumption behavior, cost-aware five-class models with data rebalancing represent a preferable solution.